

Track 39: Power Systems & Electrical Machines

June 2026

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Department: EEE | MR ROKESH TECHNOLOGY

Skill path: DC/AC Machines → Transformers → Transmission → Load Flow → Protection → MATLAB Simulation → Industrial Drives

Related: [Track 25 — PLC & Industrial Automation](#) | [Track 38 — MATLAB & Simulink](#)

Track Overview

Who this is for: EEE students targeting power utilities, industrial drives, and electrical design roles.

Prerequisites: Basic circuit theory and electromagnetism.

Tools: MATLAB/Simulink, ETAP (or PowerWorld intro), multimeter/motor lab kits, simulation software.

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Power engineering awareness	Machine characteristics report
1 Month	80–100	Junior technician level	Simulation + lab notebook
3 Months	250–300	Internship-ready	Load flow + protection study
6 Months	500–600	Power engineer entry	Industrial drive + grid study capstone

2-Week Crash Course

Days	Topics	Deliverable
D1–2	DC machines: construction, characteristics, speed control	DC motor lab/sim report
D3–4	Transformers: equivalent circuit, efficiency, regulation	Transformer calculation sheet
D5–6	3-phase induction motor, starting methods	Motor performance curves
D7–8	Power generation, transmission basics, per-unit system	Single-line diagram
D9–10	MATLAB simulation, capstone	Machine simulation project

1-Month Foundation

Week	Topics	Project
W1	DC/AC machines deep dive + lab	Machine test report
W2	Transformers, 3-phase systems, power factor	Power factor correction design
W3	Transmission lines, load flow intro (MATLAB)	Load flow simulation
W4	Protection: relays, breakers, coordination intro	Protection scheme document

3-Month Job Ready

Month	Topics	Projects
M1	Machines + transformers (advanced)	Lab + simulation portfolio
M2	Load flow, fault analysis, ETAP/PowerWorld	Grid analysis report
M3	Industrial drives, VFD, PLC integration intro, capstone	Industrial drive system capstone

6-Month Professional

Add: renewable energy integration, smart grid intro, energy audit project, ETAP certification prep, site visit documentation.

Assessment Rubric

Weekly Quiz 20% | Lab/simulation 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

High-voltage lab safety mandatory. Pair with [Track 25](#) for industrial control integration.